



Lovesick: Romantic relationships linked to opioid misuse among juveniles

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ABSTRACT

Opioid misuse (OM) is a major public health problem that is more prevalent and devastating among correctional populations, especially justice-involved adolescents (JIA). Adolescence is a critical developmental period associated with the onset of romantic relationships, substance misuse and delinquency. Social bonding theory submits that quality relationships reduce delinquency. However, prior research has not yet examined the association between relationships and OM among JIA. Adopting social bonding theory, the current study investigates two hypotheses: (H1) JIA in romantic relationships will have a lower risk for misuse relative to those not in relationships, and (H2) JIA with prosocial partners will have a lower risk for OM than those with delinquent partners. Data on 79,960 JIA from the Florida Department of Juvenile Justice were examined. OM was operationalized by past 30-day illicit or nonmedical misuse determined by urine analysis or self-disclosure. Multivariate logistic regression analyses were estimated while adjusting for covariates. Contrary to H1, relationships had higher risk for OM. Compared to JIA who were not in a romantic relationship, having a delinquent partner was associated with 2.48 higher odds of OM and having a prosocial partner was associated with 1.29 times higher odds. Intervention efforts that target youth may benefit from adopting peer models that promote prosocial relationships.

1. Introduction

The National Survey on Drug Use and Health defines opioid misuse (OM) as the nonmedical use of prescription or illicit opioids, such as oxycontin or heroin (SAMSHA, 2017). OM remains a severe public health crisis devastating millions of individuals and families across the United States (U.S.). Compelling evidence suggests that the COVID-19 pandemic has increased the risk of overdoses (Slavova et al., 2020). Additional research demonstrates that correctional populations are vulnerable to OM and significantly contribute to opioid-related overdoses and deaths (Binswanger et al., 2013; Green et al., 2018). Substance use and misuse typically begin during adolescence (Gray & Squeglia, 2018). Considering that justice-involved adolescents (JIA) are a critically underserved population vulnerable to health disparities, it is essential to identify predictors of misuse initiation among these youth. Many individuals enter romantic relationships during adolescence for the first time; social bonding theory suggests that these attachments could act as a protective factor against OM among adolescents (Dollar & Hendrix, 2015; Gómez-López et al., 2019; Laub et al., 2018).

1.1. Adolescence & opioid misuse

In 2015, roughly 5,376 adolescents died from a drug overdose, with over half of the deaths involving opioids (Curtin et al., 2017). The National Survey on Drug Use and Health found that an estimated 2.1 million people over 12 years old had an opioid use disorder (OUD), 153,000 of whom were between 12 and 17 years old (SAMSHA 2017). Populations in the juvenile and criminal justice system tend to have higher rates of substance misuse, overdoses, and related health problems relative to the general population (Binswanger et al., 2013; Bronson et al., 2017; Chassin, 2008; Sung et al., 2005; Young et al., 2007). Adolescent OM is associated with various detrimental outcomes, such as hopelessness and delinquency (Johnson & Cottler, 2020; Murphy et al., 2014; Sung et al., 2005). Furthermore, risks for relapse and overdose increase when individuals are released from correctional facilities (Binswanger et al., 2013).

1.2. Adolescent romantic relations

Romantic relationships play an important role in adolescent

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development and can affect one's well-being and health (Gómez-López et al., 2019; Sawyer et al., 2018). By age 15, many adolescents have started dating or entered a committed relationship; by age 18, most adolescents have had a romantic relationship (Carver et al., 2003; Feiring, 1996). These relationships are often a salient feature of quotidian life; romantic partners become highly embedded in adolescents' peer networks (Connolly et al., 2004). One study found that adolescents typically spend more time interacting with their romantic partners than with family or friends (Laursen & Williams, 1997). Given the centrality of romantic connections in adolescence, it is necessary to investigate how these relationships relate to risk for substance misuse and whether these patterns can inform intervention efforts.

1.3. Social bonding theory

Hirschi, an American Criminologist, first introduced the social control theory in 1969 (Hirschi, 1969). Hirschi (1969) described this theory as having four elements: attachments, commitment, involvement, and beliefs. Social control theory defines the central element, attachments, as emotional bonds that act as primary sources of socialization that occur with parents, peers, teachers, and schools (Hirschi, 1969). These bonds are facilitated or mediated by the other elements (i.e., commitment, involvement, and beliefs) (Hirschi, 1969). Laub et al. (1998) build upon Hirschi's social control theory in their social bonding theory. Social bonding theory proposes that attachments, such as romantic partners, are the foundation of strong social bonds. These attachments can prevent delinquent and unhealthy behaviors as individuals maintain these relationships (Laub et al., 1998; Laub et al., 2018). Laub et al. (1998) assert that romantic partners can serve as sources of social control, with investment in romantic relationships strengthening an individual's commitment to prosocial behaviors. These bonds may reduce the risk of substance misuse by serving as a source of support, companionship, accountability, and positive peer influence (Gómez-López et al., 2019; Laub et al., 2018).

Very few studies have investigated romantic relationships and the risk for OM, and our review did not uncover any studies that examined these relationships among adolescents. However, there is evidence that adolescent dating and romantic relationships are positively correlated with substance use (i.e., alcohol and cannabis) (Furman et al., 2007; Furman et al., 2009), with significance placed on serious romantic relationships over casual dating (Beckmeyer, 2015). Likewise, past adolescent romantic relationships may influence future substance use regardless of the relationship's continuity (Gudonis-Miller et al., 2012). Contrastingly, research among adults linked romantic relationships to a lower risk for substance misuse (Gómez-López et al., 2019; Herold, 2015), including OM (Dollar & Hendrix, 2015). In a nationally representative sample of adults from the National Longitudinal Study of Adolescent Health, Dollar and Hendrix (2015) found that married individuals were significantly less likely to misuse opioids than unmarried persons.

Cogent research underscores the importance of experiences in the relationship rather than the simple presence of one (Furman et al., 2007). Specific characteristics of romantic relationships, such as whether a partner practices prosocial or delinquent behavior, can impact substance misuse patterns (Fleming et al., 2010; Furman et al., 2007; Furman & Shaffer, 2003; Gudonis-Miller et al., 2012; Maume, 2006; Mburu et al., 2019; Meacham et al., 2013). Intimate partner delinquency, especially substance misuse, is a strong predictor of misuse and can moderate the positive effects of a quality relationship (Papp, 2010). Fleming et al. (2010) found that partner substance misuse moderated the protective effects of relationship quality on substance use (alcohol and cannabis), indicating that social bonds depend on the characteristics of the partner to whom an individual is bonded. While no prior studies have directly tested the relationship between romantic relationships and OM among adolescents, the preponderance of scientific evidence suggests that delinquent romantic relationships might

facilitate OM while prosocial relationships attenuate OM.

1.4. The current study

This study is an unprecedented investigation of romantic relationships and OM among juveniles in the criminal justice system. The findings aim to fill a critical research gap and provide insights that address a deadly crisis among a critically underserved and vulnerable population. Understanding how romantic relationships and OM transpire in JIA is crucial, given that justice-involved populations are at high risk for OM. Moreover, adolescent opioid use is a growing public health concern; romantic involvement and substance use initiation typically begins during adolescence, and romantic relationships may facilitate or attenuate substance misuse. Based on Social Bonding Theory, the current study tests two hypotheses: H1) JIA in romantic relationships will have a lower risk for misuse than those not in relationships, and H2) JIA with prosocial partners will have a lower risk of OM than those in relationships with partners who engage in delinquent behaviors. Leveraging data from the Florida Department of Juvenile Justice (FLDJJ), multivariable logistic regression is employed to investigate the association between prosocial and delinquent romantic relationships and past 30-day opioid misuse (P30D OM) among JIA. Florida has unique demographic and political features that have public health implications. The state is the most prominent political swing state in the U.S., contains the third largest population of juveniles in the criminal justice system in the U.S., and has seen a dramatic increase in opioid-related overdose deaths in the past decade (O'Donnell et al., 2017). These findings serve to advance research and intervention efforts in Florida.

2. Methods

2.1. Data

The study uses statewide data from the FLDJJ. Since 2007, the FLDJJ collects data on all minors who were arrested using a comprehensive assessment and case management process (Baglivio & Jackowski, 2013). When a minor is arrested in Florida, they complete an enrollment process to enter the FLDJJ system. As a part of the enrollment process, all youth are administered the Positive Achievement Change Tool (PACT) assessment. Trained personnel conduct semi-structured interviews using the PACT software. The interface guides all aspects of data collection; it includes open-ended questions, an interview guide, the PACT manual, and coding techniques. The PACT instrument has been validated across multiple samples of FLDJJ data in several peer-reviewed journals (Baglivio & Jackowski, 2013). This report includes 80,441 JIA in the FLDJJ PACT dataset; <1% (n = 481) of the total cases were omitted due to missing data on substance use (SU), resulting in a final dataset of 79,960 individuals.

2.2. Population

The population of 79,960 represents all youth who entered the FLDJJ from 2007 to 2015, completed the full PACT assessment, reached the age of 18 by 2015, and had data on their current drug use. The data does not include duplicate assessments of participants. Nearly 38.3% were non-Latinx White (n = 30,591), 45.6% of subjects were non-Latinx Black or African American (n = 36,443), 15.7% were Latinx (n = 12,536), and 0.5% were another race (n = 390). Roughly 21.9% of the sample were female (n = 17,497), the average age of the current sample was 14 years, and the mode was 16 years.

3. Measures

3.1. Opioid misuse

The construct of current OM is included in the PACT assessment and refers to using nonmedical prescription opioids or illicit opioids within the past 30 days. FLDJJ criteria operationalized OM as the detection through urinalysis or self-disclosure of illicit or nonmedical opioid consumption within the past-30 days, abbreviated P30D OM, at the final screen. Responses were coded into a dichotomous measure that reported whether the youth met the FLDJJ criteria for P30D OM. The response items were (0) no, did not meet FLDJJ criteria for P30D OM or (1) yes, P30D OM as indicated by urine analysis or self-disclosure. If a participant disclosed they used opioids in the past 30 days but the urine analysis was negative, the participant would meet P30D OM. Contrarily, if a participant reported they did not use opioids and the urine analysis was positive, they were classified to have used opioids in the past 30 days.

3.2. Romantic partner who engages in delinquent behaviors

The current study defines a romantic partner who engages in delinquent behaviors as dating or being sexually involved with someone who behaves in a manner considered “antisocial” or contrary to the laws and customs of society. This was operationalized by a three-item categorical measure of past-30-day relationship status. Youth were screened to determine if they were currently (within the past 30 days) in a romantic, intimate, or sexual relationship. The response options were: (0) not romantically involved with anyone, (1) romantically involved with a prosocial person, or (2) romantically involved with a person who engages in delinquent behaviors.

3.3. Control variables

Known predictors of OM (biological sex, race, age, family income, history of mental problems, history of depression, optimism, and county of residence) are adjusted for the analysis. Females in the juvenile justice system tend to have a high risk of OM relative to males (Johnson & Cottler, 2020; Shaw et al., 2019). Biological sex was operationalized by a self-reported measure (0 = male, 1 = female).

Race is a social construct that commonly includes or is used synonymously with ethnicity and national origin. Therefore, race and ethnicity refer to race. Evidence suggests that Blacks and Latinx youth in FLDJJ have much lower rates of opioid misuse relative to White youth (Cushion et al., 2019). Race was operationalized by a four-item nominal measure (0 = White, 1 = Black, 2 = Latinx, 3 = other). This variable was separated into three dummy variables: Blacks, Latinx, and Other, with Whites as the reference category, coded 0. FLDJJ restricts the release of data on age during their arrests to protect their privacy.

Age represents a proxy measure determined through the participants' date of birth and the time of their first PACT assessment. This resulted in a continuous variable ranging from 10 to 18 years of age at the time of the assessment. Prior studies using FLDJJ data indicate that opioid use is slightly younger than non-users (Johnson & Cottler, 2020). Higher household income levels are generally associated with higher odds of misuse than lower incomes. Family income was measured by a four-item ordinal variable reporting the combined annual income of the youth and family. Response options were (0) under \$15,000, (1) from \$15,000 to \$34,999, (2) from \$35,000 to \$49,999, and (3) \$50,000 and above.

Mental health history has been linked to OM (Johnson & Cottler, 2020) and may explain the relationship between toxic romantic relationships and substance misuse (Miller, 2017). The history of mental health diagnosis was measured by a dichotomous variable reporting the youth's history of mental health problems at intake. Response items were (0) no history of mental health diagnosis or (1) diagnosed with

mental health problems. In the data, depression and anxiety are combined. History of depression was measured by an ordinal variable reporting the youth's history of depression and/or anxiety at intake. Response items were (0) no history of depression/anxiety, (1) occasional depression, (2) consistent depression/anxiety but no impairment, and (3) impairment from consistent depression.

In a recent study of nearly 80,000 JIA in Florida, Johnson and Cottler (2020) found that low levels of optimism were the most impactful predictor of OM. Current optimism was measured by an ordinal variable reporting the youth's level of optimism. The measure was reverse-coded such that higher values represented lower levels of optimism. Response items were (0) high optimism, (1) normal optimism, (2) low optimism, and (3) very low optimism. High optimism indicated high aspirations, a sense of purpose, and a commitment to a better life. Normal optimism predicted normal aspirations and a sense of purpose. Low optimism showed low aspirations and a little sense of purpose or plans for the better. Very low optimism indicated that the youth believed nothing mattered and expected to be dead soon. The residing county was measured by a 68-item categorical variable reporting the county of the youth's current residence. Response options include the 67 counties in Florida and an option for out-of-state. The county data is not reported herein.

3.4. Analytical procedures

Data analysis was conducted using STATA, version 15 SE (StataCorp, 2017). A complete case analysis was appropriate given that there was minimal missing data (<1%) that was Missing Completely at Random, and the sample size was over 79,000. Demographic data were summarized using descriptive statistics. A chi-square test of independence was performed to compare whether there was a significant association between categorical variables and OM. Multivariable logistic regression was used to calculate adjusted odds ratios (aOR) and 95% confidence intervals. The covariates of biological sex, race, age, family income, history of mental problems, history of depression, current level of optimism, and county of residence were controlled in the multivariable model. OM was a limited dependent variable; therefore, the Firth approach was employed (Coveney, 2021). The Hosmer-Lemeshow test was used to confirm adequate model fit.

4. Results

4.1. Univariate analysis

In the total sample (79,960), 2.7% (2,137) met the criteria for P30D OM, 58.8% were not in a romantic relationship, 31.0% were romantically involved with a prosocial partner, and 10.2% were in a romantic relationship with a partner who engages in delinquent behaviors. Most JIA in the FLDJJ sample resided in the counties of Dade (11.3%) and Broward (8.2%).

4.2. Bivariate analysis

Table 2 reports the results from Pearson's chi-square test, which indicated that there was a significant association between relationship status and P30D OM among JIAs, $\chi^2(2, N = 79,960) = 760.75, p < 0.001$. The P30D OM group had more JIA in relationships with partners who engage in delinquent behaviors than the non-P30D OM group (28.0% versus 9.8%). The unadjusted odds ratios indicated that JIA in relationships with prosocial partners were 1.16 times as likely to meet the criteria for P30D OM as those who were not in a romantic relationship (unadjusted odds ratio: 1.16; 95% CI, 1.04–1.28) and those in delinquent relationships were 3.8 times as likely to meet criteria for P30D OM as those who were not in a romantic relationship (unadjusted odds ratio: 3.79; 95% CI, 3.42–4.22). Among the P30D OM group, 80.6% were White, 8.3% were Black, and 10.2% were Latinx. Among those

Table 1

Characteristics of justice-involved children by past-30 day (P30D) opioid misuse (OM).

	Total (n = 79,960)		No P30D OM (n = 77,823)		P30D OM (n = 2,137)	
	n	%	n	%	n	%
Intimate/dating relationship						
No relationship	47,014	58.8	46,057	59.2	957	44.8
Prosocial partner	24,751	31.0	24,170	31.1	581	27.2
Delinquent partner	8,195	10.2	7,596	9.8	599	28.0
Biological sex						
Male	62,463	78.1	61,006	78.4	1,457	68.2
Female	17,497	21.9	16,817	21.6	680	31.8
Race						
White	30,591	38.3	28,868	37.1	1,723	80.6
Black	36,443	45.6	36,266	46.6	177	8.3
Latina/o	12,536	15.7	12,318	15.8	218	10.2
Other	390	0.5	371	0.5	19	0.9
Household						
Under \$15,000	20,715	25.9	20,248	26	467	21.9
From \$15,000 to \$34,999	41,883	52.4	40,864	52.5	1,019	47.7
From \$35,000 to \$49,999	11,842	14.8	11,445	14.7	397	18.6
\$50,000 and over	5,520	6.9	5,266	6.8	254	11.9
History of mental diagnoses						
None	66,469	83.1	65,019	83.5	1,450	67.9
History of mental diagnoses	13,491	16.9	12,804	16.5	687	32.1
History of depression						
None	50,633	63.3	49,812	64	821	38.4
Occasional	21,986	27.5	21,176	27.2	810	37.9
Consistent	6,247	7.8	5,837	7.5	410	19.2
Impairment	1,094	1.4	998	1.3	96	4.5
Optimism level						
High	10,858	13.6	10,763	13.8	95	4.4
Normal	44,347	55.5	43,443	55.8	904	42.3
Low	23,856	29.8	22,781	29.3	1,075	50.3
Very low	899	1.1	836	1.1	63	2.9

Note. Data displayed as column percent.

Table 2

Pearson's chi-square test reporting intimate relationship status by past-30 day (P30D) opioid misuse (OM) status.

	Total (n = 79,960)		No P30D OM (n = 77,823)		P30D OM (n = 2,137)	
	n	%	n	%	n	%
Intimate/dating relationship						
No relationship	47,014	58.8	46,057	59.2	957	44.8
Prosocial partner	24,751	31.0	24,170	31.1	581	27.2
Delinquent partner	8,195	10.2	7,596	9.8	599	28.0

$X^2(2) = 760.75^{***}$

Note. Data displayed as column percent. *** $p < 0.001$.

assessed but did not meet the criteria for P30D OM, 37.1% were White, 46.6% were Black, and 15.8% were Latinx. For complete descriptive statistics and bivariate analysis, see [Table 1](#).

4.3. Multivariable analysis

[Table 3](#) displays the results of the multivariable logistic regression model estimating the likelihood of P30D OM while controlling for biological sex, race, age, household income, history of mental health diagnoses, history of depression, level of optimism, and residing county. Having a partner who engages in delinquent behaviors is significantly associated with current OM. JIA who reported a delinquent relationship had 2.48 times increased likelihood of OM compared to those who were not in a relationship (AOR: 2.48; 95% CI 2.20, 2.79), and those who reported prosocial relationships had 1.29 times increased likelihood of OM compared to those who were not in a relationship (AOR: 1.29; 95%

Table 3

Firth logistic regression estimating past-30 day (P30D) opioid misuse (OM).

	Model 1		Model 2	
	OR	CI	OR	CI
Intimate/dating relationship (ref = none)	–	–	–	–
Prosocial partner	1.29***	[1.16,1.44]		
Delinquent partner	2.48***	[2.20,2.79]		
Delinquent partner (ref = prosocial partner)			2.06***	[1.80,2.37]
Biological sex (ref = male)	–	–	–	–
Female	1.09	[0.98,1.22]	1.08	[0.94,1.24]
Race (ref = white)	–	–	–	–
Black	0.12***	[0.10,0.14]	0.10***	[0.08,0.13]
Latina/o	0.46***	[0.39,0.54]	0.33***	[0.27,0.40]
Other	1.08	[0.67,1.73]	0.80	[0.37,1.74]
Age in 2007	0.95***	[0.93,0.97]	0.93***	[0.90,0.96]
Income (ref = under \$15,000)	–	–	–	–
From \$15,000 to \$34,999	1.04	[0.93,1.17]	1.12	[0.96,1.31]
From \$35,000 to \$49,999	1.19*	[1.03,1.38]	1.16	[0.95,1.41]
\$50,000 and over	1.47***	[1.24,1.74]	1.53***	[1.22,1.93]
History of mental diagnoses (ref = none)	–	–	–	–
Yes, history of mental diagnoses	1.08	[0.97,1.20]	1.22**	[1.06,1.40]
History of depression (ref = none)	–	–	1.00	[1.00,1.00]
Occasional	1.59***	[1.43,1.76]	1.48***	[1.28,1.70]
Consistent	2.12***	[1.84,2.44]	1.86***	[1.54,2.24]
Impairment	2.51***	[1.97,3.21]	2.05***	[1.46,2.86]
Optimism (ref = high)	–	–	–	–
Normal	2.20***	[1.77,2.73]	2.11***	[1.55,2.87]
Low	4.56***	[3.67,5.68]	3.94***	[2.89,5.36]
Very low	7.12***	[5.02,10.08]	6.16***	[3.88,9.78]
Constant	0.02***	[0.01,0.03]	0.04***	[0.02,0.07]
Observations	79,960		32,946	

Note: aOR = Adjusted odds ratios. 95% confidence intervals (CI) in brackets. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

47,014 cases removed in Model 2 because they were not in a romantic relationship.

CI 1.16, 1.44). When comparing romantic partners with prosocial relationships as the reference category, JIA who reported delinquent relationships were 1.9 times as likely to meet the criteria for P30D OM.

4.4. Control variables

All control variables were significantly associated with current OM, except biological sex, history of mental health diagnoses, the “other” category for race, income between \$15,000–\$35,000, and certain counties of residence (county of residence is not interpreted in the multivariable models and omitted in the tables).

Compared to Whites, Blacks were 88% less likely, and Latinx were 54% less likely to misuse opioids. A one-unit increase in age was associated with a 95% decrease in chances of P30D OM. Income levels above \$35,000 were associated with a higher likelihood of OM than incomes below \$15,000. JIA with a history of consistent depression was more than twice as likely to report current OM as JIA with no history of depression. JIA with low aspirations were 4.56 times more likely to misuse opioids currently, and those who believed nothing mattered were 7.12 times more likely than JIA with high ambitions.

5. Discussion

The current study aimed to test if H1) JIA in romantic relationships will have a lower risk for misuse than those not in relationships, and H2) JIA with prosocial partners will have a lower risk of OM than those in relationships with partners who engage in delinquent behaviors. The analysis found that over half of the sample who met the criteria for P30D OM were in romantic relationships (55.2%); of those in relationships who met OM, 50.8% reported having a romantic partner who engages in

delinquent behaviors. Nearly a quarter of those who did not meet the criteria for P30D OM and had a romantic partner said their partner engaged in delinquent behaviors (23.9%). Although prosocial relationships were hypothesized to be a protective factor compared to not being in a relationship, those with prosocial relationships were 1.16 times more likely to have met OM criteria than those not in a relationship. The risk for OM was overall lower for those who did not have a romantic partner. Thus, the current study's hypotheses were unsupported by the data. Considering these findings, the simple presence of romantic relationships cannot be simplified as having negative or positive effects on an individual. Relationships involve nuanced, unique experiences and behaviors that may influence negative or positive outcomes (Furman et al., 2007). Efforts to address OM among adolescents may benefit from greater investments in research. Interventions and policies should focus on fostering healthy romantic and social relationships.

While the findings do not support the study's hypotheses, they corroborate prior studies indicating that higher substance use among adolescents in relationships (Furman et al., 2007; Furman et al., 2009); though partner characteristics are often strong predictors for substance use behaviors (Fleming et al., 2010; Furman & Shaffer, 2003; Gudonis-Miller et al., 2012; Maume, 2006; Mburu et al., 2019; Meacham et al., 2013; Papp, 2010). Consequently, the current study's findings may align with social learning theory, coined by Akers (1985), rather than social bonding (Laub et al., 1998; Laub et al., 2018), as initially posited. Ford (2008) uncovered that the nonmedical use of prescription pain relievers among adolescents was significantly associated with facets of social learning theory: peers' substance use and attitudes toward drug use, one's attitude toward drug use, and parental attitudes toward drug use.

Another study found compelling theoretical implications when investigating racial differences in risk factors for prescription OM among adolescents (Ford & Rigg, 2015). Ford and Rigg (2015) noted that social learning theory best explained OM among White adolescents; social bonding theory best explained OM among Latinx adolescents; parents and parental bonds served as protective factors against OM among Black adolescents. Thus, the presence of a relationship alone may not capture the nuances in the association between adolescents' romantic relationships and OM. The current study found that JIAs in a romantic relationship were more likely to misuse opioids than those not in a romantic relationship. A multitude of factors may contribute to this finding. Future research should examine relationships' association with OM and other factors that mediate this association.

5.1. Public health implications

Juvenile justice institutions have become de facto behavioral health systems, especially for adolescents from disadvantaged backgrounds (Kumm et al., 2020). National data shows that the criminal justice system is the second largest referral source (second only to self-referrals) for publicly funded substance misuse treatment programs, accounting for over 30% of all treatment admissions and nearly 40% of opioid-specific admissions (SAMSHA2015). Improved access to high-quality, evidence-based substance misuse treatment services in the justice system will be critical to addressing the opioid crisis. The National Institutes of Health sponsored the Justice Community Opioid Innovation Network to study approaches to increase high-quality care for people with OM and OUD in justice settings. Despite evidence that not being in a relationship was associated with lower risk, over 40% of juveniles were involved in romantic relationships. Adolescents should be encouraged to pursue healthy social and romantic partnerships and be equipped with the skills of the agency to avoid, identify or terminate unhealthy relationships. Rather than discouraging romantic relationships, stakeholders could include training on developing healthy relationships in their drug education and awareness programs, particularly among romantically involved adolescents. Many adolescents across racial/ethnic backgrounds are subject to romantic partners who engage in delinquent behaviors. JIA can benefit from programs that effectively teach them

how to avoid, detect, and terminate unhealthy romantic relationships. It may also be effective to develop opioid awareness and treatment strategies for adolescent couples, especially in high-risk areas.

5.2. Limitations

The study's limitations should be considered in the findings and implications. First, only youth who had completed full assessments were included in the study. This may indicate that participants represent higher-risk youth, and the finding may not be generalized to all youth who come into contact with the juvenile justice system. Second, cross-sectional data limited the ability to establish the temporal sequence. Data on relationships over time and at different points can provide a thorough understanding of adolescent romantic relationships on opioid use and misuse within the juvenile justice system. Third, the study primarily relies on a self-report assessment and assumes that the adolescent sample can distinguish a relationship as delinquent or prosocial. Considering the perspectives of parents or guardians, teachers, and counselors may offer an objective interpretation of the relationship. Accordingly, this includes the absence of the second partner reporting on the relationship. Future studies should capture valuable information from all relevant stakeholders to capture a holistic and potentially less biased relationship representation. Additionally, the sample represents Florida JIA; thus, the findings may not be generalizable to JIA in other states. Analyzing nationally representative samples may uncover important nuances and enable interventions to target at-risk youth nationwide. Moreover, OM included heroin and prescription opioid misuse, which prevents an analysis of the potential differences between each drug type. Although opioid use was a limited dependent variable, with less than three percent of youth reporting past 30-day opioid use, the Firth approach indicated that the limited variation was not problematic. Comprehensive data on the severity of OM and the degree of delinquency of the romantic partner can improve the current research. Lastly, the construct of romantic partners does not distinguish between sexual activity and romantic love; prior research suggests that distinct concepts can have conflicting effects on delinquent behavior (McCarthy & Casey, 2008).

5.3. Recommendations

Future research should replicate the current study using longitudinal data to identify the temporal sequence, distinguishing between prescription and illicit opioids, and specific measures of relationship quality and partner characteristics. The moderating effects of biological sex, race, and partner substance misuse should be explored. Moreover, future research should replicate the current study using gender as a comparison. Nationally representative samples should be leveraged to improve external validity. Intervention efforts should consider fostering positive romantic bonds to precipitate treatment uptake.

6. Conclusion

The prevalence of OM among adolescents in the criminal justice system remains dangerously high. Theory and research underscore the importance of romantic relationships in shaping and influencing adolescent behavior, with implications for substance misuse. This study was an unprecedented examination of the relationship between romantic relationships and OM among adolescents in the juvenile and criminal justice system. Findings both corroborate and contradict the social bonding theory. Adolescents in romantic relationships with partners who engage in delinquent behaviors were nearly 2.5 times as likely to meet the criteria for misuse relative to those not in a relationship. Given that relationships are critical during this stage of development, prevention and intervention efforts may benefit from integrating peer-to-peer models of adolescent substance misuse prevention that consider romantic relationships.

CRediT authorship contribution statement

Micah E. Johnson: Conceptualization, Methodology, Software, Validation, Writing – review & editing. **Sashawn D. Lawrence:** Conceptualization, Software, Writing – review & editing. **Elzbieta K. Wiedbusch:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Data Transparency

The research reported in this publication was approved by the Institutional Review Boards at the University of South Florida (STUDY001497) and the Florida Department of Juvenile Justice (FLDJJ). The data are not publicly available. Request for these data must be made directly to FLDJJ. Several articles have been published using these data. However, the authors are not aware of any research that has used these data to investigate the specific relationships examined in the present study.

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